How well do a journal's recent acceptance times predict the next paper's? An out-of-time evaluation across 800 journals

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Abstract

Background: Authors, funders and evaluators use a journal's reported or past time to acceptance as a forecast of how long a new submission will take, but such statistics have not been tested out of time, and claims that a journal accepts papers 'within X days' are rarely checked.

Methods: We used publisher-deposited received and accepted dates for 1,737,272 research articles in 800 Springer Nature, Wiley and Taylor & Francis journals (781 evaluable). At forecast origins in January 2023 and January 2024, and in a rolling quarterly panel, forecasts built only from papers already visible were scored against the 298,469 papers submitted in the following six months, using pinball loss, interval coverage and event-probability scores. Inference used journal-cluster and imprint-cluster bootstraps. The received date was checked against preprint posting dates.

Results: Journal identity explained 26.5% (95% CI 24.7%–28.3%) of the variance of log acceptance time. Journal-specific forecasts lowered loss by 15.7% relative to publisher-pooled forecasts, but 80% intervals spanned about five-fold in days and covered 77.0% of later papers (journal average; 76.4% of papers; imprint-cluster CI −4.0 to −2.0 points around nominal). The shortfall was absent under random splitting (−0.0 points) and in a stationary simulation, indicating non-exchangeability between cohorts. Lognormal, shrinkage and learned forecasters did not beat empirical quantiles of the last 12 months. A quantile-level recalibration learned on earlier cohorts restored coverage (+1.4 points) without loss in accuracy. 12.0% of journal-origins accepted at least half of their papers within 72 days (9.5%–12.5% across robustness variants) and 1.6% at least 80%. In Springer Nature journals with Research Square preprints, 95% of received dates were no later than 14 days after the preprint was posted, supporting the received date as the original submission date; this could not be tested for Wiley and Taylor & Francis.

Conclusions: A journal's last year of acceptance times is a useful but wide and slightly over-confident forecast. Reports of editorial-process statistics should give quantiles, cohort definitions and uncertainty, and 'within X days' claims should not be read as guarantees. Findings are conditional on acceptance, on three publishers and on publisher-defined dates.

Keywords: peer review; publication delay; time to acceptance; probabilistic forecasting; calibration; editorial process; research reporting

Background

Journals and publishers advertise speed, and authors, funders and evaluators repeat such statements when choosing where to publish or when setting expectations about time to acceptance. Whether the statistics behind these statements—usually a median or average of past papers—are a usable forecast of the experience of the next paper has not, to our knowledge, been tested. The question is a matter of reporting integrity in editorial processes: statistics that are defined inconsistently or presented without uncertainty can mislead the people who rely on them [1, 2, 3].

Empirical work has described how long peer review and publication take. In a stratified sample of 2,700 papers from 135 journals, most of the variance in submission-to-acceptance time lay among articles within a journal [4]. Studies have documented trends and determinants of delay in economics, accounting, medicine and across PubMed [5, 6, 7, 8, 9, 10], a COVID-19 speed-up [11], a strain on publishing capacity [12], and the fact that the definition and reporting of review times differ across journals and from article-level values [1, 2, 3]. Stage-wise models of one editorial workflow exist [13]. A registered-style plan for a PubMed-wide analysis intends to predict article-level acceptance lag with a random training/test split [14]; it states that it was not preregistered, and no results were available when we searched. Commercial tools already present historical medians as planning aids; we found no validation of them.

What we did not find is an evaluation of journal-level turnaround statistics as forecasts: scored against later outcomes, using only information visible when the forecast is made, and reporting calibration. We ask: (RQ1) how much of the variance of acceptance time lies between journals, between years and between articles; (RQ2) how accurate and calibrated are forecasts built from a journal's visible past, and which simple estimator is best; (RQ3) why do such forecasts miscalibrate; (RQ4) how often is a claim that a journal accepts a given share of papers within 72 days supported, and are such claims reliable out of time; (RQ5) do features available around submission add information; and (RQ6) is the publisher-deposited received date a valid measure of original submission? RQ1 extends earlier work to a larger and more recent dataset; RQ2–RQ6 are, as far as a non-systematic search could establish, new. The study is empirical meta-research and makes no causal claims.

Methods

Design, pre-specification and order of events

The first-cycle protocol (frame, cleaning rules, cohorts, forecasters, metrics, seven contrasts and five hypotheses) was committed to the project's git history before any forecast error was computed; this is a pre-specification by the analyst, not a registered report, and no external time stamp exists. In a second

cycle, an independent review identified weaknesses and further experiments were added. Their order is disclosed: the received-date probe, a recalibration experiment and a learned-forecaster experiment were designed in code and run once before their protocol text was written; the dependence, data-quality and calibrated-probability experiments and the follow-up panel, null/simulation, robustness and article-level experiments were specified and committed before being run. All amendments, post-hoc analyses and failed or mis-specified thresholds are logged in the repository (deviations D1–D11 and cycle-2 log). Seven of the first-cycle hypotheses were evaluated as pre-specified; results that did not support them are reported.

Data

Article-level received and accepted dates are deposited in Crossref by Springer Nature, Wiley and Taylor & Francis. For each publisher we took the 1,000 ISSNs with most journal articles published in 2023, merged print and online ISSNs (1,684 journals), and kept journals in which at least 80% of 40 random 2023 records carried both dates and the estimated number of dated 2023 articles was 150–3,000 (802 entries; two ISSN pairs were not merged, leaving 800 distinct journals with data). All records published from 1 January 2021 to 20 October 2026 were downloaded. The estimand is D, the number of days from received to accepted, conditional on acceptance. Records were cleaned with rules fixed in advance: duplicate DOIs, D<1 or >1,460 days, acceptance more than seven days after first online publication, and non-research items by title or fewer than five references. Of 1,829,907 dated records, 1,737,272 remained; median D was 124 days. Journals were assigned to an imprint by the modal DOI prefix of their articles (Springer, BioMed Central, Nature, Wiley, Blackwell, Taylor & Francis, other).

Validity of the received date

Published articles that Crossref links to a preprint were sampled (up to 30 per journal; 6,829 articles in 388 journals; the collection was interrupted and uses the part written). Research Square 'In Review' preprints are posted when a manuscript enters review at a Springer Nature journal; if the received date is the original submission date, received minus posting date should be near zero or slightly negative, and rarely large and positive. We report the share with received no later than 14 days after posting and the share more than 90 days after posting. Wiley and Taylor & Francis articles with Research Square preprints are independent postings and cannot validate their dates.

Forecasting task

Origins were 1 January 2023 and 1 January 2024 (Figure 1). At an origin the author's information set contains papers that were first online before the origin and already received and accepted. The target cohort comprises all papers received in the following 182 days. A journal-origin was eligible with at least

100 visible papers in the preceding 12 months and 50 target papers. Forecasters of the distribution of ln D were evaluated at 19 quantile levels: F0, empirical quantiles of the last 12 months (Hazen definition [15]); F1, the same over 24 months; F2, publisher-pooled quantiles ignoring journal identity; F3, a lognormal fitted by maximum likelihood; F4, an empirical-Bayes lognormal [16, 17]; F6, exponentially time-decayed empirical quantiles (half-life 182.5 days). Two cycle-2 forecasters used earlier cohorts only: R, F0 with quantile levels recalibrated by the empirical distribution of probability integral transform values of earlier target cohorts (fitted on 2022 targets for the 2024 origin and on 2022–2023 targets for 2025), and L, a gradient-boosted quantile regression [18] on journal-level features (past quantiles, size, trend, volume growth, publisher).

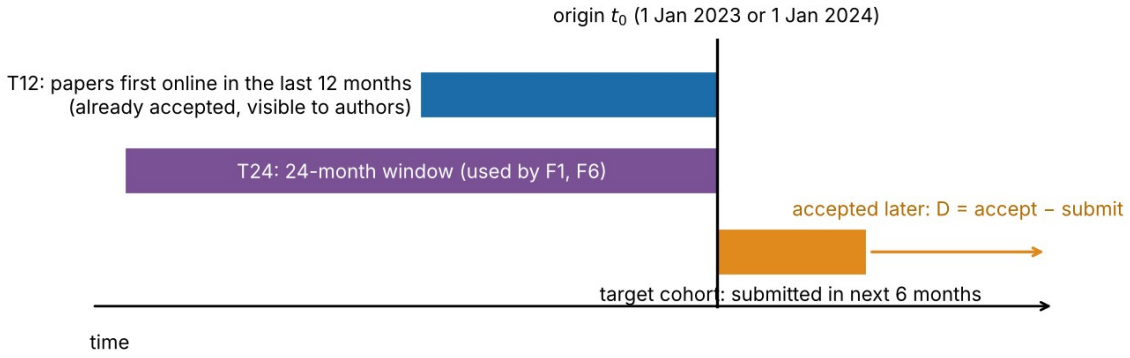


Figure 1. Conceptual schematic (not data) of the out-of-time protocol. Forecasts use only the visible training set and are scored against papers submitted after the origin.

Scoring and inference

The primary score is the mean pinball loss over quantile levels on the ln-day scale [19, 20]; secondary scores are coverage of central intervals, the interval score [21], Brier scores for the events D ≤ 72, 120 and 180 days, and probability-integral-transform histograms [22]. Journals are the unit of analysis. Because journals of one imprint are correlated and the two origins overlap, uncertainty is reported with journal-cluster bootstraps [23, 24] and, for the main results, with imprint-cluster bootstraps, across-imprint tests, stratified bootstraps, leave-one-publisher-out estimates and, for the rolling panel, a two-way (imprint × origin) bootstrap. Paired comparisons use Wilcoxon signed-rank tests [25] with Holm correction; shares use Clopper–Pearson intervals [26]. Variance components of ln D (journal, journal × year, residual) were estimated by method of moments from acceptance years 2021–2025.

Additional analyses

A rolling panel used quarterly origins from January 2022 to April 2024, 91-day target cohorts and a common follow-up cap of 480 days. Two null worlds tested the cause of any under-coverage: an

exchangeable benchmark (random split within a journal's complete submission cohort at the observed sample sizes) and a generative simulation without drift in which arrivals grew at each journal's observed rate, durations were drawn from the journal's own target distribution, and the visibility rule was applied. Robustness variants excluded records with fewer than 10 or 15 references, records accepted on days with batch acceptances, journals with at least 10% of papers accepted within 14 days, and 75 journals flagged for fast-lane or implausibly short durations. A strictly out-of-time article-level model was trained on 2022 targets and tested on 2024 targets with author count, received month and weekday (with or without reference count), against a control with the journal quantile only.

Use of AI

The analysis code and drafts of this manuscript were produced with an AI system (Claude, Anthropic) acting as an autonomous research assistant in a sandboxed environment; separately launched instances of the same model family performed adversarial reviews, a prior-art search and a reproducibility audit. No human expert has reviewed the work. The authors must verify the analyses and take responsibility for the content.

Results

Data and validity of the received date

The cleaned data comprised 1,737,272 articles; 22.8% were accepted within 72 days. Forensic checks found little heaping of D at multiples of 30 days (ratio 1.00), 1.2% of papers with $D \leq 7$ days and an expected weekday pattern. Of 4,222 articles with a Research Square preprint, 4,018 were in Springer Nature journals (212 journals): the median received date was 14 days before posting, 95% were received no later than 14 days after posting and 2.6% more than 90 days after (Figure 2). Journals whose medians looked fast (30 journals, 391 articles; median D 70 days) showed the same pattern (93% and 4.6%). The pre-set criterion that at least 70% of articles fall within ±14 days of posting was not met (47%) because these preprints are posted about two weeks after receipt; the criterion was mis-specified and the two alternative statistics are post hoc. For Wiley and Taylor & Francis journals the probe is uninformative (their Research Square preprints are independent postings: median received 79 and 100 days after posting).

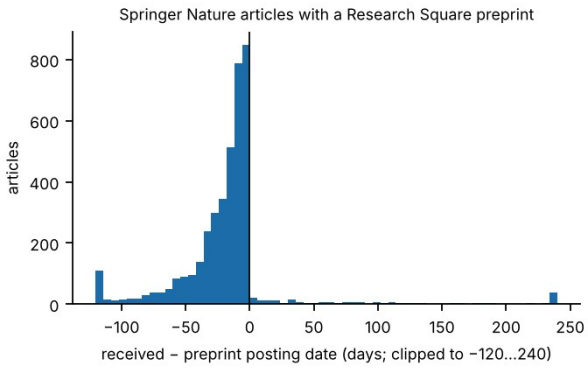


Figure 2. Received date minus Research Square preprint posting date for Springer Nature articles (days; clipped). Mass at and below zero is consistent with the received date being the original submission date.

Variance structure (RQ1)

Between-journal differences explained 26.5% of the variance of ln D (95% CI 24.7%–28.3%), journal × year differences 4.4% and residual between-article variation 69.1%. Leaving out one publisher at a time gave journal shares of 24.5%–29.0%; excluding 75 flagged journals gave 26.1%; the article-type variants gave 25.9%–27.4%. Thirty non-frame PubMed journals gave 27.4% (wide CI 17.2%–34.5%).

Forecast accuracy (RQ2)

The evaluation covered 1,531 journal-origins from 781 journals and 298,469 target papers. Journal-specific forecasts clearly beat the publisher-pooled forecast (F0 − F2 pinball difference -0.0376; imprint-cluster CI -0.0419 to -0.0304; skill 15.7%; same sign in 7/7 imprints). Differences among the journal-specific forecasters were below 2% of loss (Tables 1 and 2). F0 outperformed the lognormal F3 and empirical-Bayes F4 in all imprints, F6 was marginally better than F0 in all imprints, and the apparent advantage of the 12-month over the 24-month window (F1 − F0) was not robust to imprint clustering (p = 0.07). Additional forecasters tested by an independent reviewer (skew-normal, pooled-shape and blended) and our learned forecaster L were no better than F0 (L − F0 +0.0002, 95% CI -0.0002 to +0.0007). The lognormal forecasters also overstate the chance of acceptance within 72 days.

Table 1. Out-of-time forecast performance, 1,531 journal-origins (781 journals).

Forecaster	Mean pinball (ln-days)	Skill vs F2	80% interval coverage − 80 (points) [journal-cluster 95% CI]	Mean interval width (ln-days)
F0 empirical, 12 mo	0.2019	15.7%	−3.0 [−3.5, −2.6]	1.60
F1 empirical, 24 mo	0.2027	15.4%	−3.1 [−3.6, −2.6]	1.61
F2 publisher-pooled	0.2395	0.0%	−0.4 [−1.3, +0.5]	1.91
F3 lognormal	0.2047	14.5%	+0.8 [+0.3, +1.3]	1.76
F4 empirical Bayes	0.2044	14.6%	+1.0 [+0.5, +1.5]	1.75
F6 time-decayed	0.2011	16.0%	−2.8 [−3.3, −2.4]	1.60

Table 2. Pairwise differences in pinball loss with journal-cluster and imprint-cluster intervals.

Contrast (loss A − loss B)	Mean difference	Journal-cluster 95% CI	Imprint-cluster 95% CI	Across-imprint t-test p	Imprints with same sign
F4 − F0	+0.0025	[+0.0020, +0.0031]	[+0.0009, +0.0051]	0.022	7/7
F3 − F0	+0.0028	[+0.0022, +0.0034]	[+0.0011, +0.0054]	0.018	7/7
F1 − F0	+0.0008	[+0.0003, +0.0013]	[+0.0001, +0.0012]	0.071	6/7
F6 − F0	-0.0008	[-0.0010, -0.0005]	[-0.0009, -0.0006]	0.002	7/7
F0 − F2	-0.0376	[-0.0414, -0.0339]	[-0.0419, -0.0304]	0.000	7/7
F4 − F6	+0.0033	[+0.0027, +0.0039]	[+0.0018, +0.0057]	0.007	7/7
F4 − F3	-0.0003	[-0.0003, -0.0002]	[-0.0004, -0.0002]	0.005	7/7

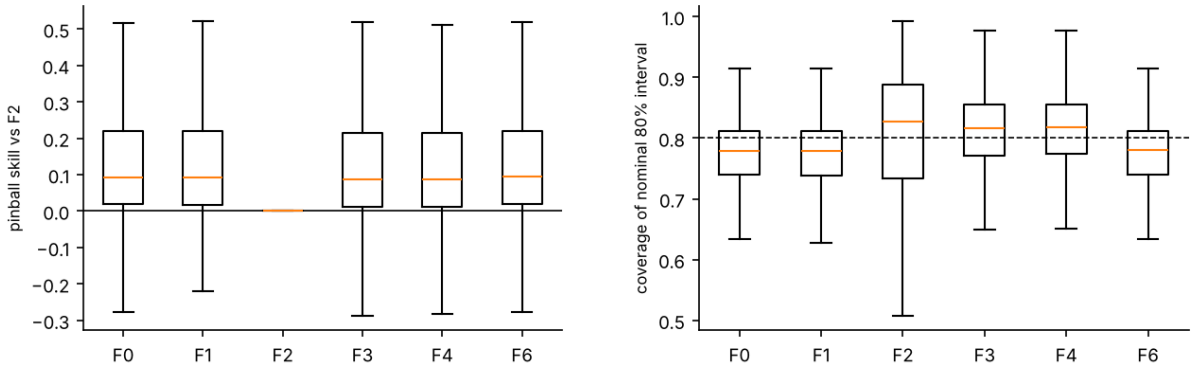


Figure 3. Left: pinball skill of each forecaster relative to F2, by journal. Right: out-of-time coverage of nominal 80% intervals by journal (dashed line = 0.80).

Calibration and its causes (RQ3)

Nominal 80% intervals from F0 covered 77.0% of later papers when averaged over journals and 76.4% when averaged over papers; the shortfall of −3.0 points had a journal-cluster CI of −3.5 to −2.6, which widened to −4.0 to −2.0 with imprint clustering and was −3.6 paper-weighted (journal-cluster CI −4.3 to −3.0). Under the pre-specified rule (a CI excluding ±3 points) the verdict on H2 is therefore inconclusive, not negative. The shortfall differed by imprint, from −5.6 points (largest: BioMed Central) to −1.0 points, but was negative in all 7 imprints. In the rolling panel with a common follow-up of 480 days (7,417 journal-origins, 787 journals) it ranged from −1.5 to −3.4 points across ten origins without a monotone trend, was −2.2 points overall (two-way CI −3.4 to −1.3), and the 95% predictive interval for a future origin was −4.0 to −0.8 points (Figure 4). Skill relative to F2 in the panel was 16.2%.

Neither finite-sample variability nor a stationary process explained the shortfall (Figure 5, Table 3): random splitting inside a complete submission cohort gave −0.0 points (CI −0.2 to +0.2), and a simulated world with the observed volume growth, the visibility rule and no drift gave +0.1 points (CI −0.1 to +0.2), against −3.0 observed. Using the complete submission cohort of the preceding year, which an author could not know, still under-covered (−1.5 points), and an independent reviewer found that the backward oracle did so too, so the residual reflects non-exchangeability between cohorts in both time directions rather than

a one-way slowing. Using only recently submitted papers that have already been accepted was far worse (−12.9 points). Journal-level shortfalls were not predictable out of time (R^2 0.001), and their spread (SD 8.4 points) was well above binomial sampling variation (3.2 points).

Table 3. Coverage shortfall of the empirical 80% interval under different information sets and null worlds.

Information set / world	Coverage − 80 (points)	95% CI (journal-cluster)
Visible set (observed)	−3.0	[−3.5, −2.6]
Complete submission cohort (oracle)	−1.5	[−1.9, −1.2]
Random split of the complete cohort	−0.0	[−0.2, +0.2]
Simulated world without drift	+0.1	[−0.1, +0.2]
Submission cohort truncated at origin	−12.9	[−13.7, −12.2]

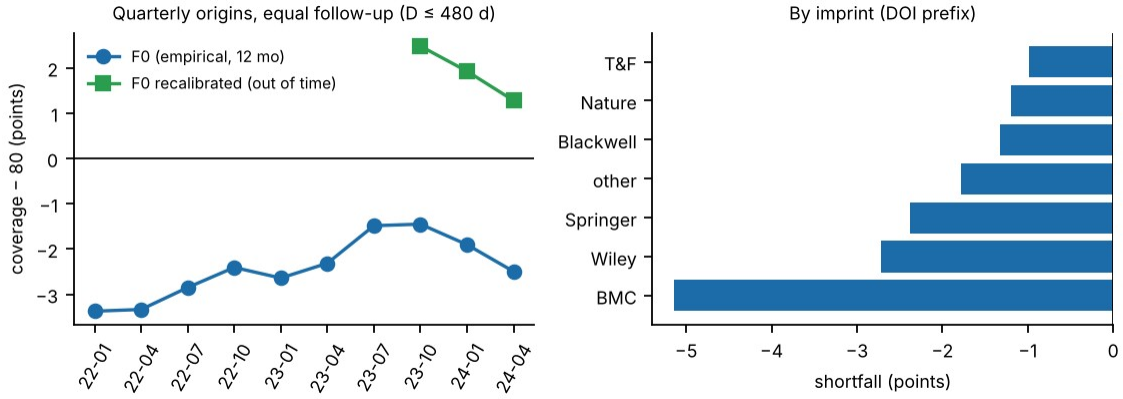


Figure 4. Left: coverage shortfall at quarterly origins with equal follow-up, for F0 and for F0 recalibrated out of time (recalibration evaluable at three origins). Right: shortfall by imprint.

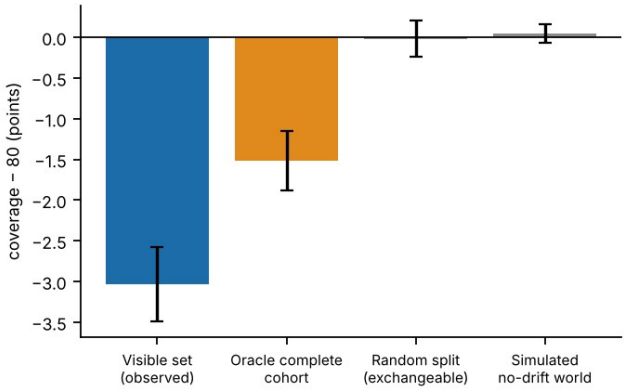


Figure 5. Shortfall under four information sets or worlds (journal-cluster 95% CIs).

A quantile-level recalibration learned only on earlier cohorts (nominal 0.10 and 0.90 map to 0.092 and 0.928) moved the 2024–2025 coverage from −2.5 to +1.4 points (CI +0.9 to +1.8) with no significant change in loss (-0.00008, CI -0.00033 to +0.00016). In the panel, a recalibration learned from early origins

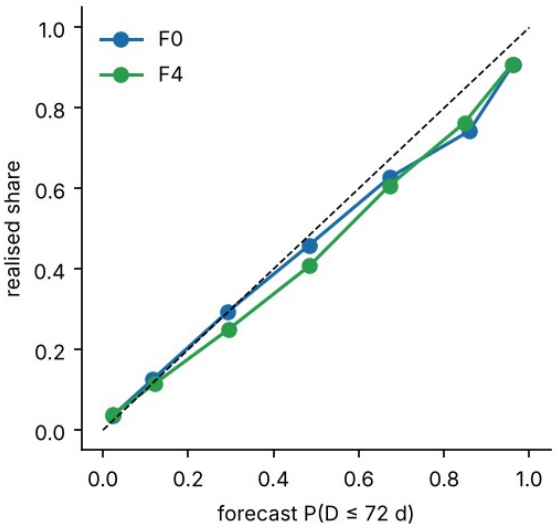
over-corrected (+1.8 points; CI +0.2 to +3.1) because the shortfall was larger in the early origins. The correction is therefore a partial remedy whose size should be re-estimated regularly.

Drift (exploratory)

The mean squared year-to-year change in a journal's median ln D was 7.3 times the value expected from sampling noise under a permutation null; because the top 5% of journal-year pairs carry 53% of the sum of squares, a robust (MAD-based) ratio of 5.3 is a safer summary. The median absolute change was 0.09 ln-units, and 18.8% of 3,105 pairs changed by more than 0.2 ln-units.

How often is a '≤ 72 days' claim supported? (RQ4)

The realised share of target papers accepted within 72 days was at least 0.5 in 184 of 1,531 journal-origins (12.0%; journal-cluster CI 9.9%–14.2%), at least 0.8 in 25 (1.6%; 0.9%–2.4%) and at least 0.9 in 7 (0.5%). The 72-day threshold is illustrative of maximum-time criteria; 120 and 180 days gave 48% and 80% of journal-origins reaching 0.5. Excluding batch records, fast-lane journals or low-reference articles, or all of these together, left the share between 9.5% and 12.5%; excluding 75 flagged journals gave 8.9%. A substantial part of this share is therefore due to journals with a fast lane for some article types. Fast journals were commonest, by coarse title-keyword coding (exploratory), among chemistry, materials and physics (28% of 99) and clinical medicine journals (23% of 195), and rare among engineering and computing journals (7% of 68). Past evidence overstated future speed at the fast end: for the 23 journal-origins (19 journals) forecast at least 0.8 (mean forecast 86%) the realised share was 75%. Calibrating event probabilities with the same map lowered the Brier score slightly (differences -0.0011 to -0.0003; CI excluded zero in 3 of 6 origin-threshold combinations) and brought the reliability slope for the 72-day event from 0.90 to 0.97 (2024 origin) and from 0.84 to 0.88 (2025 origin).



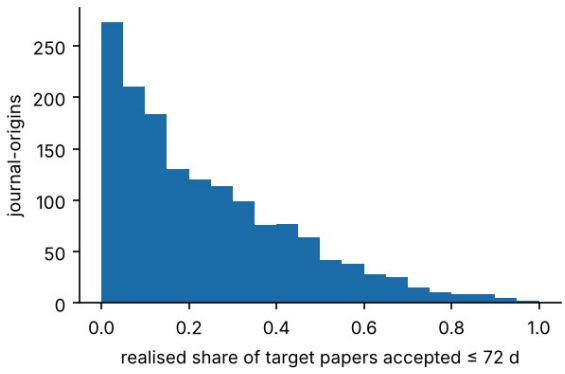


Figure 6. Left: reliability of forecast P(D ≤ 72 d) (F0, F4). Right: distribution of the realised share of papers accepted within 72 days.

Article-level information (RQ5)

In a strictly out-of-time model (trained on 2022 targets, tested on 158,735 papers of 2024 in 766 journals), author count, received month and weekday improved pinball loss by 1.0% over the control (95% CI 0.6%–1.4%); journal-demeaned features gave 0.8%. Adding reference count, which is not known before submission, raised the gain to 3.0% (2.3%–3.6%), concentrated in the fast tail (τ = 0.10: 5.8%; τ = 0.90: 0.5%), consistent with reference count identifying short article types rather than anything knowable at submission.

Robustness and external check

Table 4. Robustness of the headline quantities to article type, batches, flagged journals and single origins.

Variant	Journals	F0 shortfall (points) [CI]	Skill vs F2	Share of journal-origins with ≥ 50% ≤ 72 d	Journal share of variance
Base	781	−3.0 [−3.5, −2.6]	15.7%	12.0%	26.5%
References ≥ 10	777	−3.1 [−3.5, −2.6]	15.7%	11.1%	26.8%
References ≥ 15	769	−3.2 [−3.7, −2.7]	15.9%	11.0%	27.4%
No batch-accepted records	774	−2.9 [−3.4, −2.5]	15.9%	12.5%	26.4%
No fast-lane journals	733	−3.1 [−3.5, −2.6]	15.2%	9.5%	25.9%
Combined	714	−3.0 [−3.5, −2.5]	15.8%	9.9%	27.2%
Excluding 75 flagged journals	708	−3.0 [−3.5, −2.6]	15.7%	8.9%	26.1%
Single origin 2023	765	−3.3 [−4.0, −2.8]	15.4%	–	–
Single origin 2024	766	−2.7 [−3.3, −2.1]	16.2%	–	–

Leave-one-publisher-out estimates of the F0 shortfall ranged from −3.5 to −2.5 points. Crossref and PubMed dates agreed for 99.997% of 59,885 shared DOIs, which shows only that two channels carrying the same publisher data agree. In 58 journal-origins from 30 eligible PubMed journals outside the frame (a non-random set), journal-specific forecasts beat the pooled forecast by 14.8%, but the shortfall (−0.7 points, CI −3.6 to +2.1) was compatible with both zero and the main result; this is not a replication.

Table 5. Hypotheses and outcomes (H1–H7 pre-specified in cycle 1; X1–X3 specified in cycle 2, see Methods for order of events).

ID	Hypothesis	Outcome	Estimate
H1	Journal share of variance < 50%	Supported	26.5% [24.7%, 28.3%]
H2	F0 80% coverage differs from 80% by > 3 points	Inconclusive	journal mean −3.0; paper-weighted −3.6; imprint CI [−4.0, −2.0]
H3	Shrinkage (F4) beats F0 in small journals only	Not supported (small-n regime untestable)	F4 worse than F0 in all strata and imprints; F7/F8, learned L no better
H4	< 5% of journal-origins have ≥ 50% accepted ≤ 72 d	Not supported	12.0%; variants 9.5%–12.5%; flagged-excluded 8.9%

H5	Article covariates improve loss by < 2%	Specification-dependent	pre-submission features 1.0% (supported); with references 3.0% (not supported)
H6 (expl.)	Drift exceeds sampling noise	Supported	robust ratio 5.3
H7 (expl.)	12-month window beats 24-month window	Not robust	F1 − F0 +0.0008; imprint p = 0.07
X1	Received ≈ original submission (Research Square)	Supported for Springer Nature; threshold as set not met; untestable elsewhere	95% ≤ +14 d; 2.6% > +90 d
X2	Recalibration brings coverage within ±1.5 points at no loss	Supported at 2024–25; over-corrects in the panel	+1.4 points; panel +1.8
X3	Learned forecaster no better than F0	Supported	+0.0002 [-0.0002, +0.0007]

Discussion

Three findings stand out. First, a journal's own recent history is informative but modestly so: it cuts forecast loss by about 16% relative to ignoring journal identity within a publisher, yet 69% of the variance lies between articles, so 80% intervals span roughly a five-fold range of days. The variance partition agrees with the earlier finding that most variance lies among articles within a journal [4] on a much larger and more recent dataset, and is not claimed as new. Second, simple forecasts are over-confident by about two to three points of coverage, and the cause is not statistical noise or a stationary process but non-exchangeability between cohorts: intervals built from one period fail to contain a similar share of papers from the next, whichever the direction, and the instability varies unpredictably across journals. A one-parameter recalibration from earlier cohorts removes most of the average bias without loss of accuracy but over-corrects when the bias is shrinking, so it should be re-estimated routinely. Third, the simplest estimator was hard to beat: lognormal, shrinkage and learned forecasters did not improve on empirical quantiles of the last 12 months, in line with the left-skewed, heavy-tailed shape of within-journal log durations (median skewness -0.79).

For authors, evaluators and publishers the practical message is about reporting. Medians alone hide a wide and drifting distribution; statements of the form 'X% of papers are accepted within T days' should name the cohort (accepted or submitted), the period, whether desk rejections and article types with fast lanes are included, and should give quantiles with uncertainty. Phillips and Horn found inconsistent reporting among health-policy journals [1], and our results quantify how much a single past statistic can mislead: even among the minority of journals that accepted most papers within 72 days, realised shares fell short of past-based forecasts. These are statistics about published, accepted papers only, and they should not be read as guarantees or as probabilities that a submission is accepted.

The study complements descriptive analyses of publication delay and the planned PubMed-wide model [10, 14, 27] by testing forecasts out of time, across three publishers and all fields, and by reporting what would have to change in practice: recalibration, quantiles and explicit cohorts.

Limitations

- Conditioning on acceptance: only accepted papers have dates; rejected and withdrawn submissions, which may differ in duration, are unobserved, so results do not describe the time or chance of a given submission being accepted.
- Date semantics: the received date is publisher-defined. It is validated only for Springer Nature journals through Research Square preprints (a self-selected subset); Wiley and Taylor & Francis dates, and the finding that a minority of journals accept most papers within 72 days, rest on unverified definitions. Crossref–PubMed agreement does not verify them. Attempts to compare with publisher article pages were blocked.
- Frame and generalisation: three publishers, journals with about 150–3,000 dated articles per year, 56% Springer Nature among evaluated journals; Elsevier, IEEE and others are outside the frame; the external check (30 journals, non-random) is inconclusive. The frame was defined from 2023 volumes, and eligibility requires at least 50 target papers.
- Inference and design: two confirmatory origins (ten in an exploratory panel), overlapping information sets, and an imprint structure that inflates journal-cluster precision; imprint-cluster intervals are wider and, with seven clusters, approximate. Several tests are post hoc or exploratory and the explanation of the shortfall is characterised rather than identified.
- Article types and batches: durations mix article types and fast lanes; our variants reduce but do not remove this, and article type is not observed.
- Remedy: the recalibration map is learned from few earlier cohorts, was evaluated at three origins in the panel and over-corrected there.
- Prior work and novelty: the literature search was not systematic and most full texts were not read; the claim of novelty is therefore stated as 'we found no'. No human reviewer has examined the work.
- Scope: the study analyses statistics about editorial processes, not peer review itself.

Conclusions

A journal's acceptance times over the last year are a useful but wide, slightly over-confident forecast for its next papers. Simple empirical quantiles are as accurate as more elaborate forecasters, a one-parameter recalibration improves average coverage, and 'within 72 days' claims are supported for a minority of journals and overstate future speed. Turnaround statistics should be reported as drifting distributions with explicit cohorts and uncertainty.

Abbreviations

ICC, intraclass correlation (journal share of variance); D, days from received to accepted; CI, confidence interval; F0–F6, R, L, forecasters defined in Methods; Q1, first JCR quartile (not asserted for any journal here).

Declarations

Ethics approval and consent to participate: not applicable; the study analyses public article metadata.

Consent for publication: not applicable.

Availability of data and materials: raw Crossref and PubMed records, processing code, configuration and all results are in the project repository (https://github.com/Bhargavteja-9779/rp_c_8, branch claude/autonomous-research-institute-nz8co5), which is not yet archived with a persistent identifier; a DOI-bearing deposit (for example Zenodo) must be created before submission. Live API data change over time; SHA-256 hashes of the cleaned files used are in results/data_manifest.json.

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Note: journal titles are not abbreviated according to Index Medicus because the style guide could not be applied automatically; the authors should abbreviate them.